

* Unit :- 5 Memory Management

⇒ It is a core function of managing allocating & deallocating the system's primary memory (RAM) among running processes.

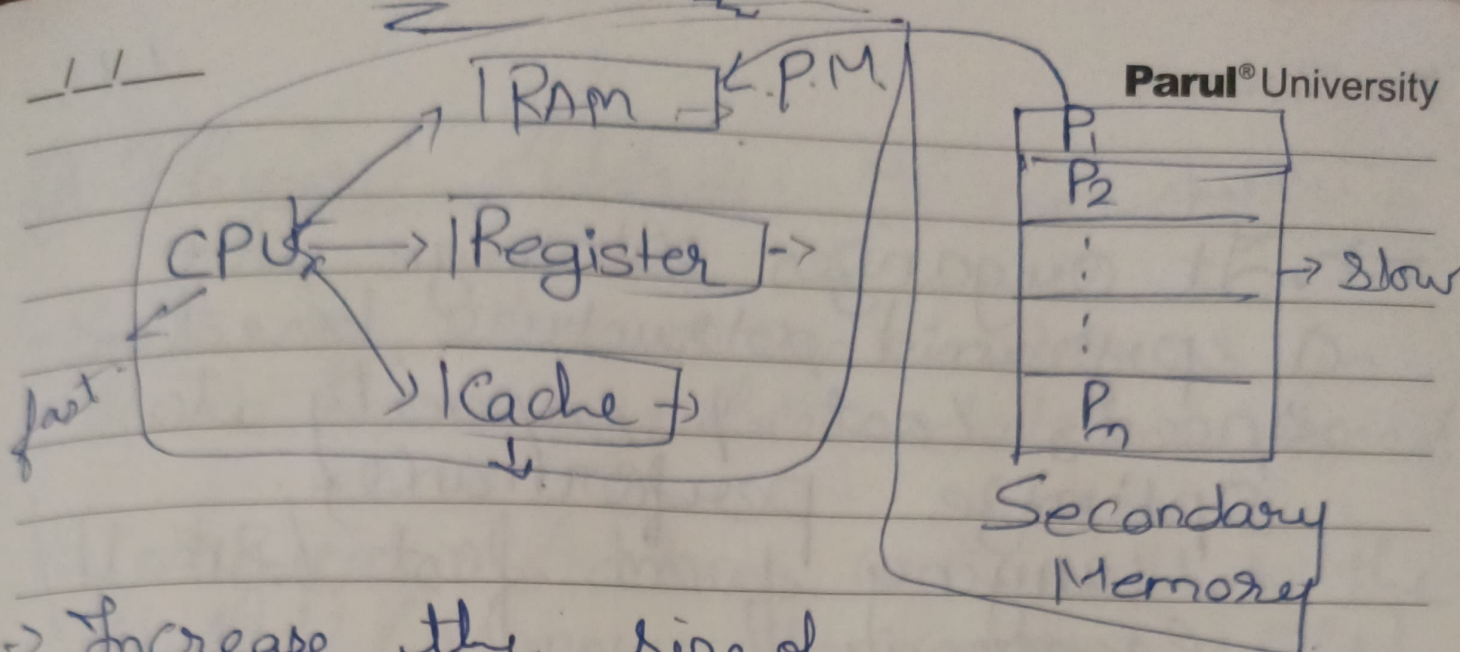
⇒ It ensure efficient CPU utilization, track memory usage & protects process isolation,

Key Responsibilities of M.M :-

-> Tracking :- Keep records of which memory locations are in use & which are free.

-> Allocation & Deallocation :- Allocating memory to processes when needed & freeing it upto completion.

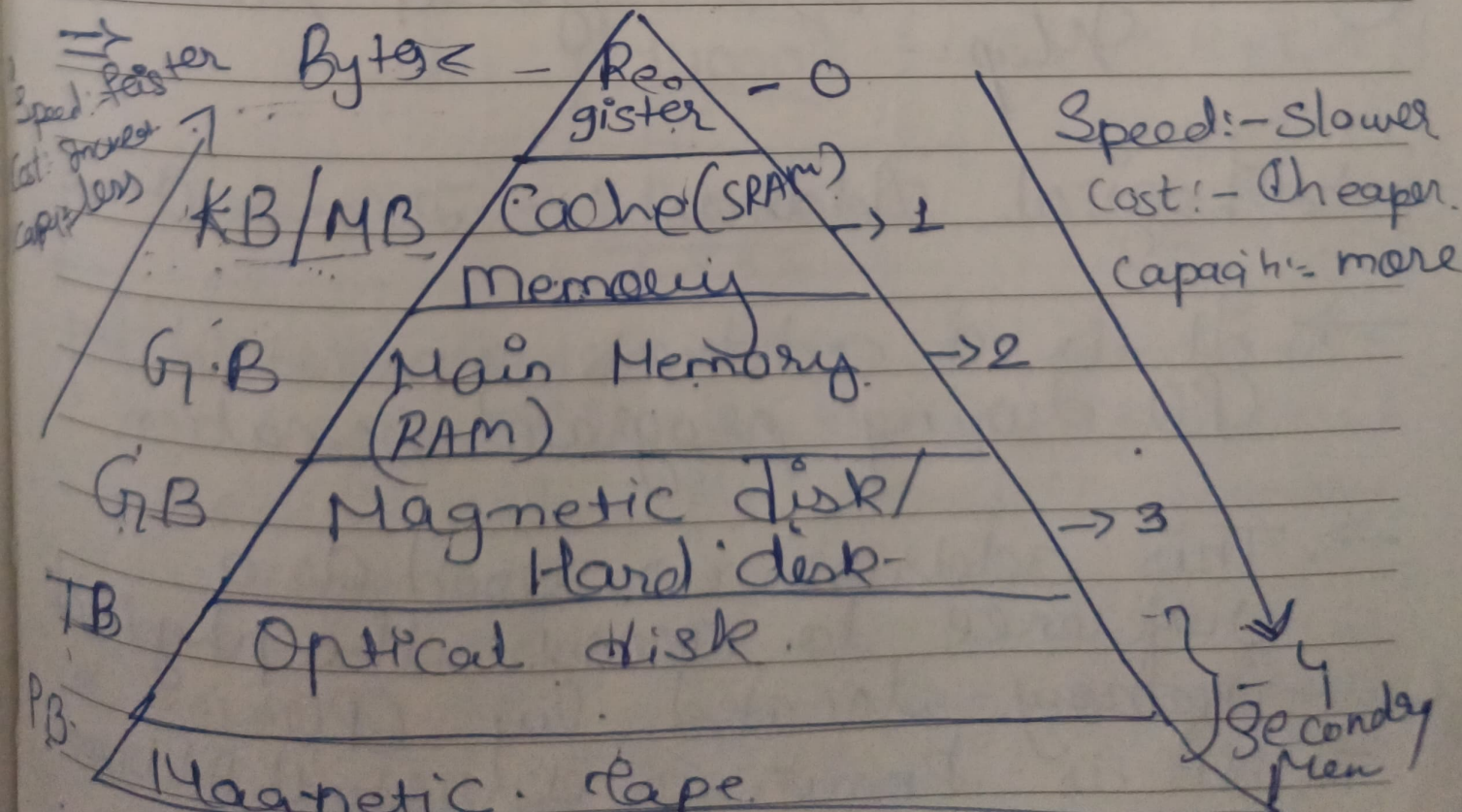
-> Isolation & Protection :- Preventing processes from accessing memory space belonging to others, ensure



→ Increase the size of RAM it will directly affect the cost of the system.

↓ Method of managing P.M.
This is known as degree of multiprogramming.

→ Memory Hierarchy :-



⇒ It organizes storage into a pyramid structure based on speed, cost & capacity to optimize performance ranging from fast / small / expensive.

→ It is the memory organizer of a particular system to balance its overall cost & performance.

⊛ Logical & Physical Address Map :-

⇒ Logical Address :- It can be changed.

→ It is an address generated by CPU during program execution.

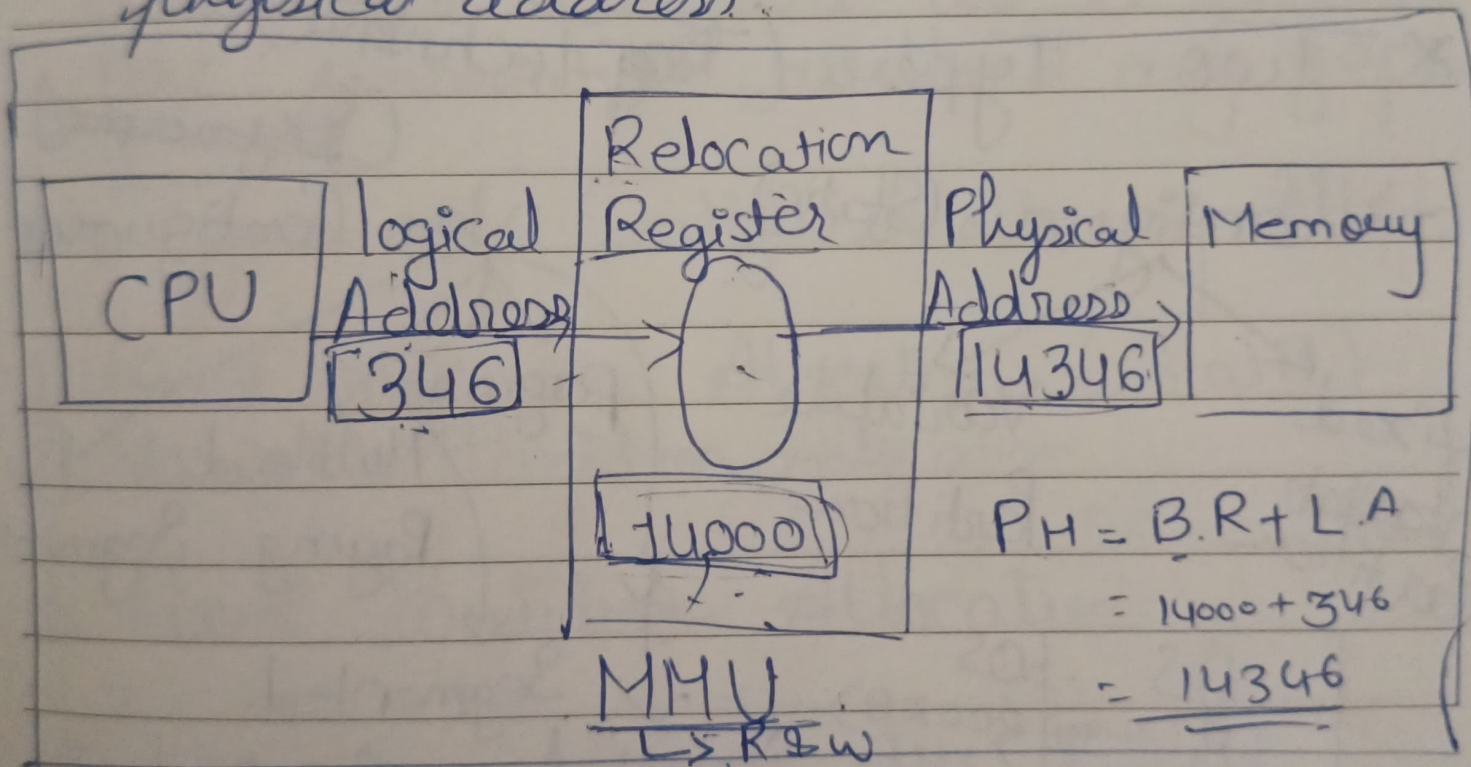
→ This address is used as a reference to access the physical memory located by CPU.

→ It is known as a Virtual Address.

① Physical Address:- It can't change.

→ It is the actual address in Main Memory where data is stored

→ Physical Address are used by the M.M.U (Memory Management Unit) to transform logical address to physical address.



→ Relocation Register:- This hardware register holds the base physical address where the process's memory begins.

* MMU:- It maps logical & physical

* Memory Allocation :-

⇒ It is the process of assigning a block of memory in RAM to a program, process or variable so that it can store & manipulate data during execution

* Two Types :- (~~Two~~ Technique)

→ Contiguous (Static)

fixed ~~partition~~ Partition Variable Partition

	OS	OS
P ₁	(100-150)	(200 KB)
P ₂	(200-300)	Process
P ₃	(350-600)	150 KB
P ₄	↓	300 KB
		free

(~~Dynamic~~) Non-Contiguous

Paging

Multilevel

Paging

Segments

Segmented

Paging

Inverted
Paging

①. Contiguous Memory Allocation -

=> It is a M.M technique where each process is allocated a single contiguous block of memory.

=> This means that all memory addresses assigned to a process are consecutive, without any gap.

* Types of Contiguous Memory Allocation

1. Fixed Partition Allocation (Static)
2. Variable Partition Allocation.

1. Fixed Partition Allocation:-

-> No. of partitions are fixed

-> We divided the M.M into partitions & the size of each partitions can be same/equal or can be different also.

- The partitions can't overlap
- A process must be contiguously present in partition for the execution.

	OS		OS	⑤
0	P₁ ^{1MB} 1MB	2MB	P ₁	8MB
1	P ₂	8MB 4MB	P ₂	8MB
2	P ₃	8MB	P ₃	8MB
3	P ₄	8MB	P ₄	8MB
4	P ₅	16MB	P ₅	8MB
	M.M		M.M	

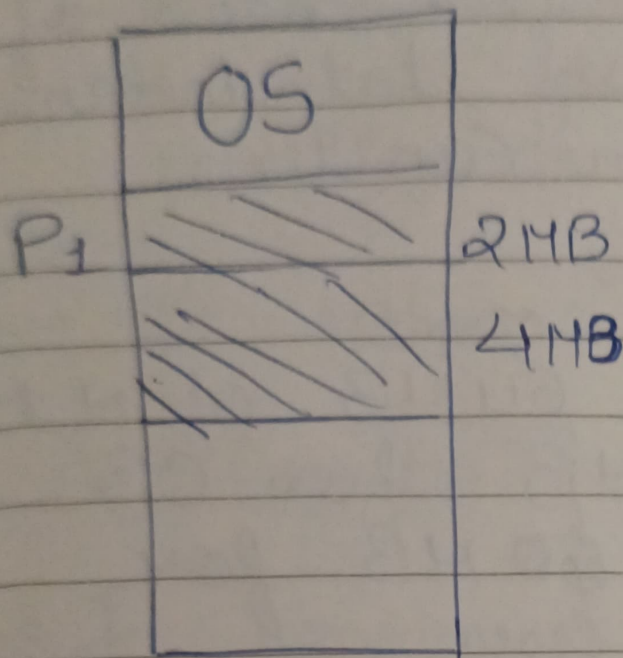
$P_1 = 1MB$, $P_2 = 2MB$

→ If the process is 1MB then the 1MB is of partition is wastage which is known not utilized by other process is known as Internal Fragmentation

* Problems / Limitations:-

- Internal fragmentation
- limit in process size
- limitation on degree of multiprogramming
- External fragmentation.

* Variable Partition (Dynamic)



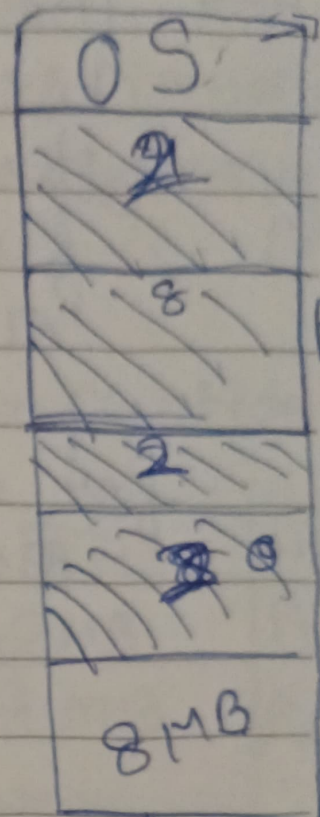
P₁ = 4MB

P₂ = 8MB

P₃ = 20MB

P₄ = 8MB

P₅ =



⇒ In the variable Partition ^{Mem.} Allocation, the partition size is not declared initially. It is declared at the time of process loading.

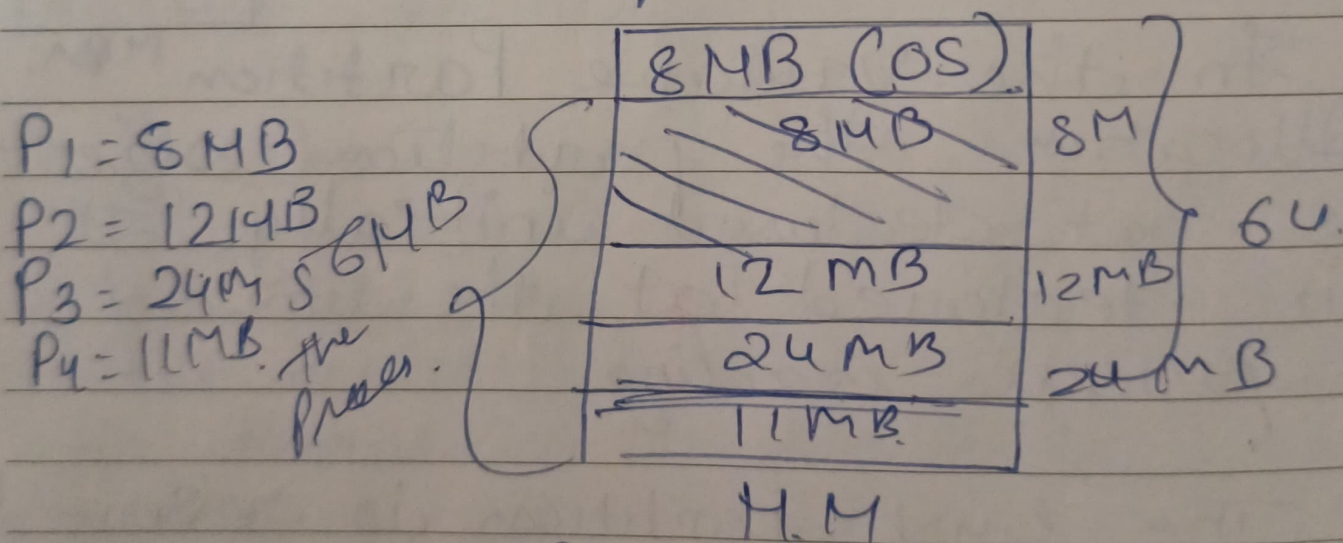
→ The first partition is reserved for the OS. & the remaining space is divided into parts.

→

→ The size of each partition will be equal to the size of the process.

⇒ The partition size varies according to the need of the process so that internal fragmentation can be avoided.

for eg:- Total ^{50 MB} 60 MB of M.M
out of ~~50 MB~~ 60 MB for OS
& other ~~50 MB~~ 60 MB for the processes.



⇒ If ~~P₂~~ ^{& P₅} is removed then other process is come of 16 MB ~~4 MB~~ then (16 MB) is remain empty.

⇒ So, it is known as External fragmentation.

⇒ External Fragmentation means memory becomes more & more fragmented & memory utilization declines.

E.g. we will

→ It solves using Compaction.

Ex. Compaction is :-

→ In Contiguous A process R Can't span between 2 different different location.

OS	
2MB	P ₁
8MB	P ₂
2MB	P ₃
20MB	P ₄
8MB	P ₅

① Compaction

R

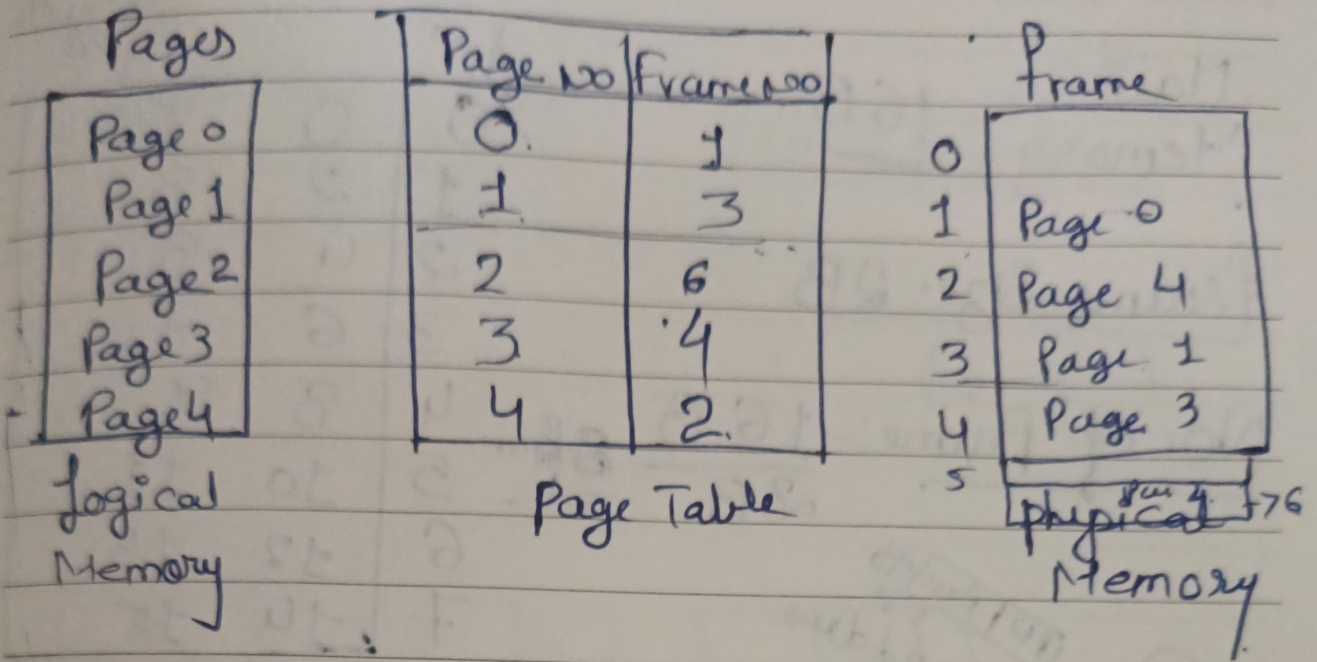
⇒ Compaction is an OS M.M technique that rearranges fragmented, non-contiguous free memory into one large block by shifting running processes.

⇒ It is time consuming & wastes CPU time.

⇒ Basically it is undesirable but still we can use it.

* Paging :-

- The M.M is partitioned into equal fixed-sized chunks & that each process is also divided into small fixed size chunk of the same size.
- The chunks of the process are called pages. (Logical M)
- The Chunks of a memory are called frames. (Physical Mem)
- The OS maintain a page table for each process.
- It contains the frame location for each page in the process.
- Paging is to break a program into smaller fixed-sized block.



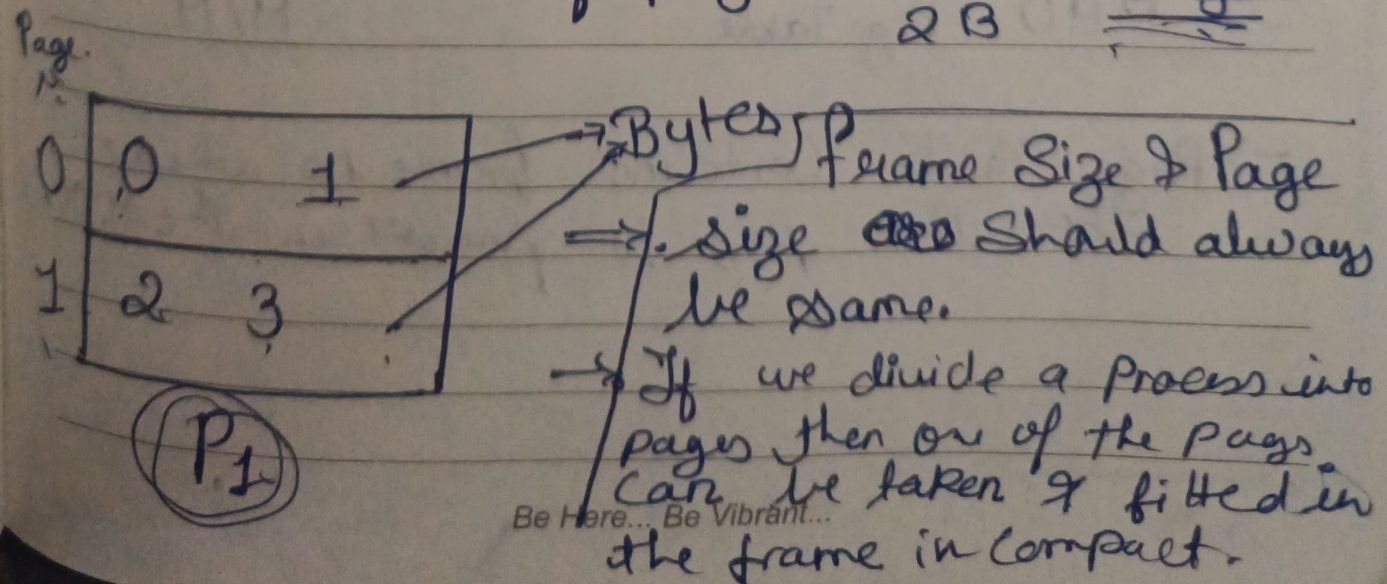
→ Process 1 size is 4 bytes & page size is 2 bytes.

→ Total Size of the process is 4 bytes & single page is 2 bytes.

$$P.S = 4 B.$$

$$\text{Page Size} = 2 B.$$

$$\text{No of pages} = \frac{4 B}{2 B} = \underline{\underline{2 \text{ bytes}}}$$

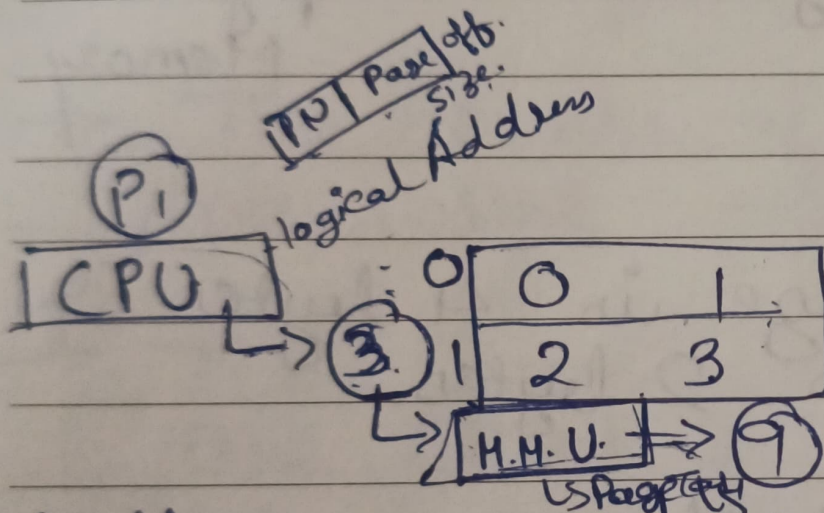


Main Memory = 16 B.

Frame Size = 2 B

No. of frame = $\frac{16 \text{ B}}{2 \text{ B}} = \underline{8 \text{ frames}}$

Frame:	bytes	
0	0	1
1	2	3
2	4	5
3	6	7
4	8	9
5	10	11
6	12	13
7	14	15



M/M.

⇒ Here we need Mapping.

⇒ The CPU generated is not Absolute address.

⇒

MMU → Page Table

0	0	1	File
1	2	3	File
2	4	5	PO
3	6	7	File
4	8	(9)	P1
5	10	11	File
6	12	13	
7	14	15	

M/M.

→ Page Table Contains frame no. where the that page is present in the H.M.